

Operating System

操作系统

课程目标

- 通过本课程的学习，使学生了解并掌握操作系统的概念、理论、方法、技术，掌握操作系统的实现方法
- 使学生体会到其他专业基础知识在操作系统中的综合应用，理解具体操作系统的原理，加深对专业基础知识的理解
- 使学生能够利用所学的理论知识解决实际问题，培养学生具有对计算机操作系统及相关信息领域实际复杂工程问题、采用科学有效的方法进行研究的能力
- 对英文技术资料快速阅读和准确理解的能力，以及能够运用英文进行科技交流和沟通的能力。

具体目标

- 课程目标1：掌握计算机操作系统基础知识及原理，培养能够对计算机操作系统相关的设计、性能等问题进行比较、分析和定量计算的能力。
- 课程目标2：培养采用科学方法，针对计算机操作系统及相关信息领域的复杂问题进行文献调研和分析的能力。
- 课程目标3：培养针对计算机操作系统及相关信息领域的技术问题和研究目标，选择合适的研究路线，设计相应的实验方案的能力。
- 课程目标4：培养针对计算机操作系统领域的复杂问题，构建实验系统，开展实验，对实验结果进行综合分析，得到合理有效的结论的能力。
- 课程目标5：熟悉计算机操作系统及相关信息领域的英文专业术语，培养对于相关英文技术文献和标准的快速阅读和准确理解能力，能够运用英文专业术语进行沟通的能力。

Mathematics Discrete Mathematics Probability

Data Structures Operating System

Compiler Principle

Database Systems Software Engineering

Physics Digital Logic and Digital System

The computer organization and architecture

Computer Network

Textbook

- 《Operating System Concepts》（Seventh Edition）Abraham Silberschatz et al，高等教育出版社，2007年3月
- 操作系统概念（原书第10版）Abraham Silberschatz et al，郑扣根 译，机械工业出版社，2023年

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Part Four Storage Management

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■ Preparation

■ Take notes

■ Review

■ Practice

■ Q&A

Operating System

- Homework (programming)
- midsemester
- final

考核方式

- 课程的考核环节主要包括：平时作业、实验、期中考试和期末考试课程
- 其中平时成绩（包括作业和实验）占总成绩的**20%**
实验包括进程管理、内存管理实验，实验由学生独立设计、实现，并完成实验报告。
- 期中考试成绩占总成绩的**20%**
- 期末成绩占总成绩的**60%**。

Chapter 1: Introduction

- What is an Operating Systems
- Computer-System Organization
- Computer-System Architecture
- Operating-System Structure
- Operating-System Operations
- Process Management (进程管理)
- Memory Management (内存管理)
- Storage Management (存储管理)
- Protection and Security
- Systems and Computing Environments
(系统和计算环境)

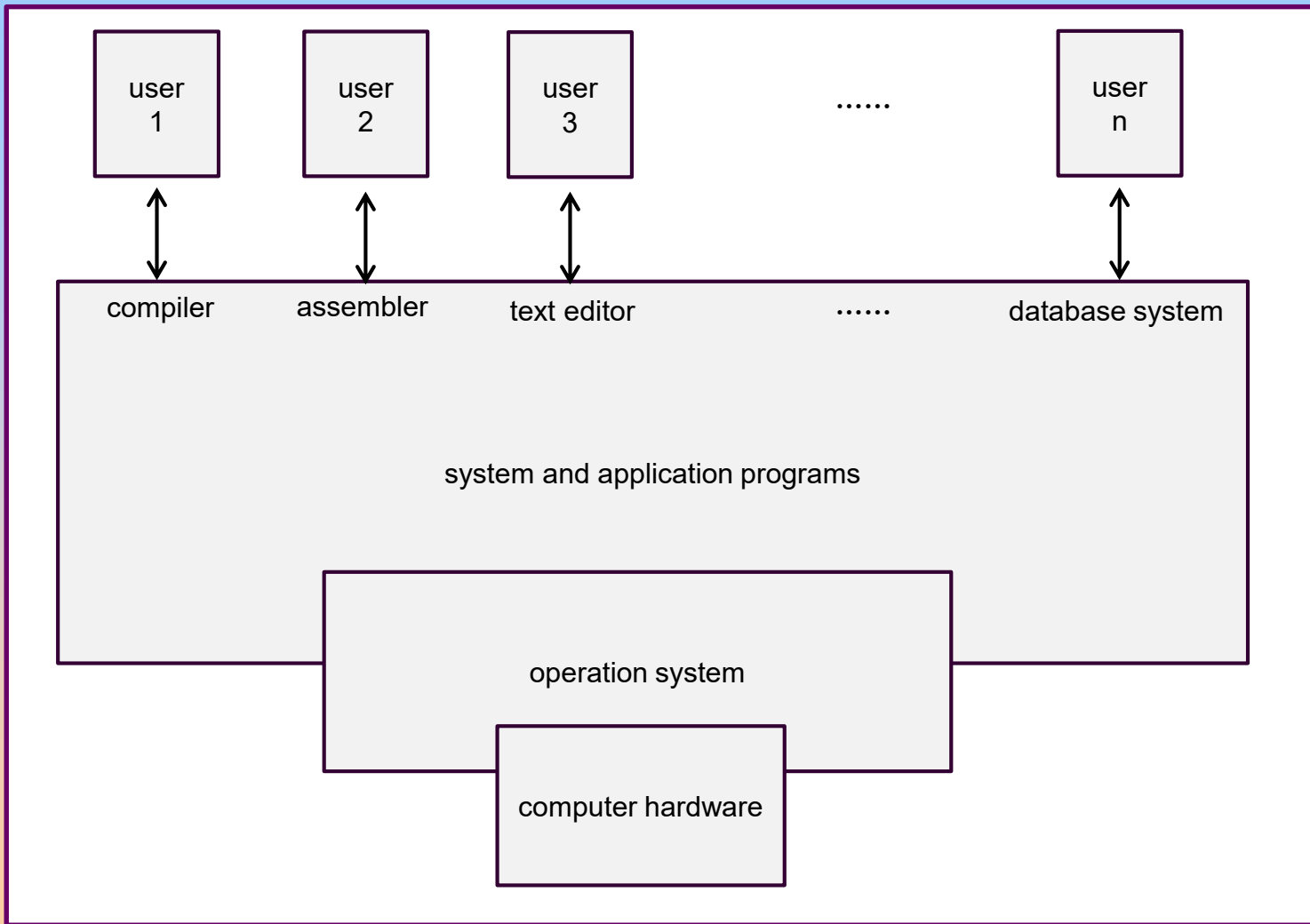
What is an Operating System?

- A program that acts as an intermediary between a user of a computer and the computer hardware.
- Operating system goals:
 - ☞ Execute user programs (用户程序) and make solving user problems easier.
 - ☞ Make the computer system convenient to use.
- Use the computer hardware in an efficient manner.

Computer System Components

1. Hardware (硬件) – provides basic computing resources (CPU, memory, I/O devices).
2. Operating system – controls and coordinates the use of the hardware among the various application programs for the various users.
3. Applications programs – define the ways in which the system resources are used to solve the computing problems of the users (compilers, database systems, video games, business programs).
4. Users (people, machines, other computers).

Abstract View of System Components



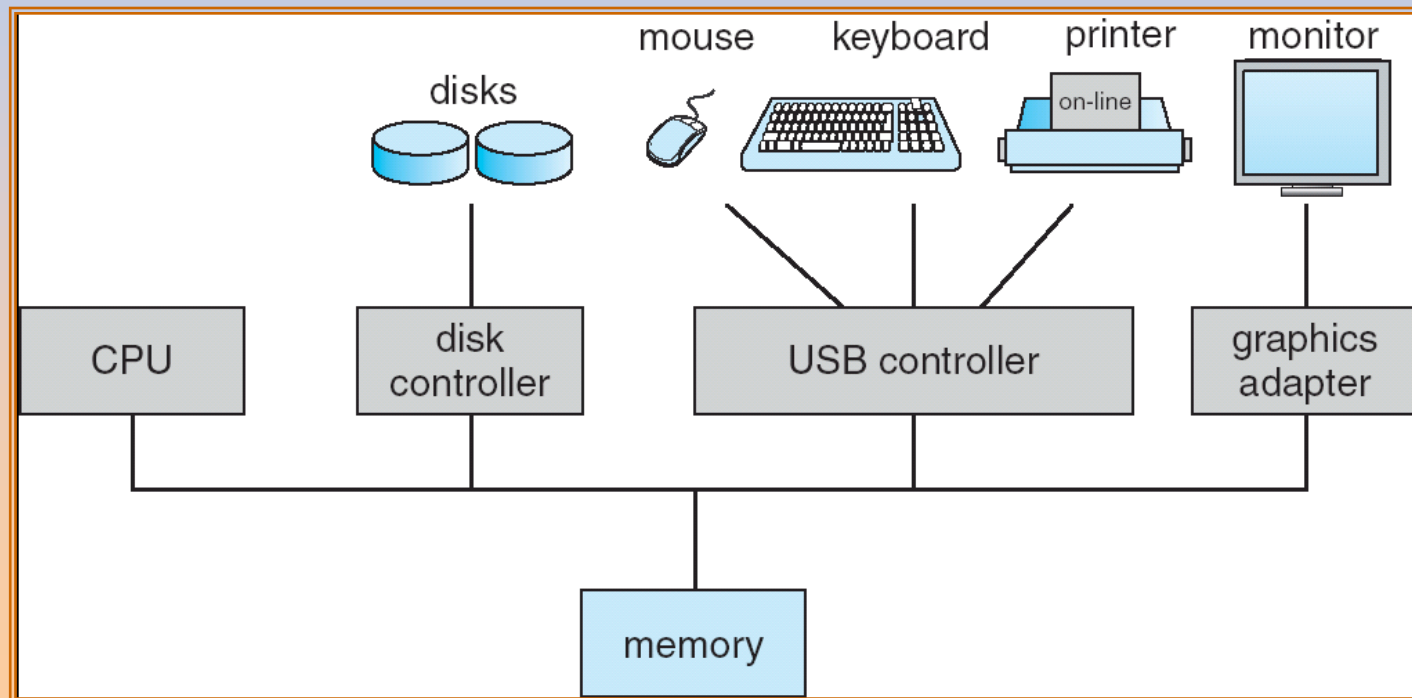
Operating System Definitions

- Resource allocator(资源分配器) – manages and allocates resources.
- Control program(控制程序) – controls the execution of user programs and operations of I/O devices .
- Kernel(内核) – the one program running at all times (all else being application programs).

Computer System Organization

■ Computer-system operation

- ☞ One or more CPUs, device controllers(设备控制器) connect through common bus(总线) providing access to shared memory
- ☞ Concurrent execution of CPUs and devices competing for memory cycles



Computer-System Operation

- I/O devices and the CPU can execute concurrently.
- Each device controller is in charge of a particular device type.
- Each device controller has a local buffer(缓冲区).
- CPU moves data from/to main memory(内存) to/from local buffers
- I/O is from the device to local buffer of controller.
- Device controller informs CPU that it has finished its operation by causing an *interrupt*(中断).

Common Functions of Interrupts(中断)

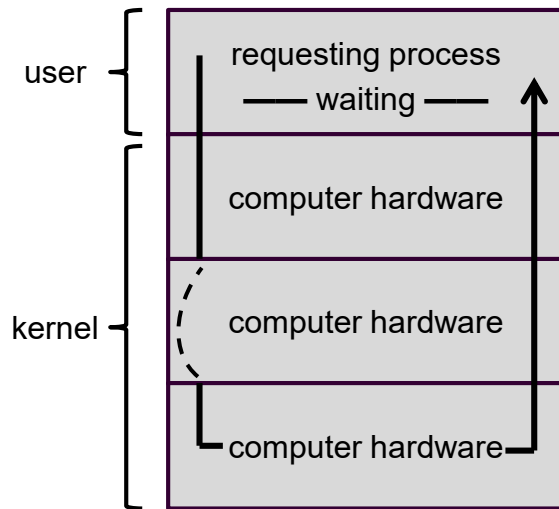
- Interrupt transfers control to the interrupt service routine generally, through the *interrupt vector*, which contains the addresses of all the service routines.
- Interrupt architecture must save the address of the interrupted instruction.
- Incoming interrupts are *disabled* while another interrupt is being processed to prevent a *lost interrupt*.
- A *trap*(陷阱) is a software-generated interrupt caused either by an error or a user request.
- An operating system is *interrupt* driven.

I/O Structure(I/O结构)

- After I/O starts, control returns to user program only upon I/O completion.
 - ☞ Wait instruction idles the CPU until the next interrupt
 - ☞ Wait loop (contention for memory access).
 - ☞ At most one I/O request is outstanding at a time, no simultaneous I/O processing.
- After I/O starts, control returns to user program without waiting for I/O completion.
 - ☞ *System call(系统调用)* – request to the operating system to allow user to wait for I/O completion.
 - ☞ *Device-status table* contains entry for each I/O device indicating its type, address, and state.
 - ☞ Operating system indexes into I/O device table to determine device status and to modify table entry to include interrupt.

Two I/O Methods

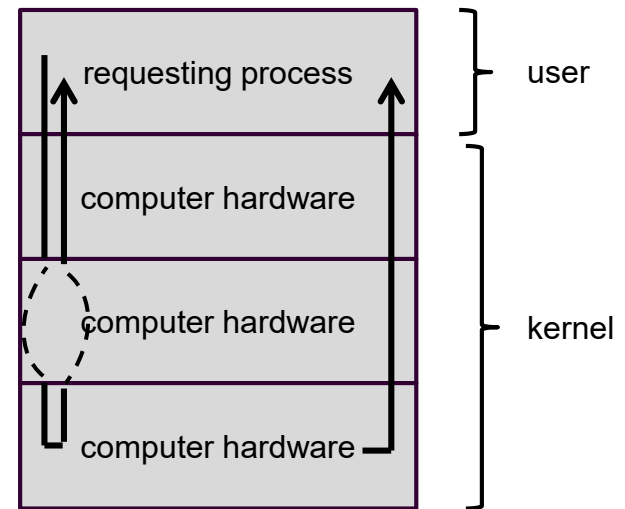
Synchronous



time →

(a)

Asynchronous



time →

(b)

Direct Memory Access Structure

- Used for high-speed I/O devices able to transmit information at close to memory speeds.
- Device controller transfers blocks of data from buffer storage directly to main memory without CPU intervention.
- Only one interrupt is generated per block, rather than the one interrupt per byte.

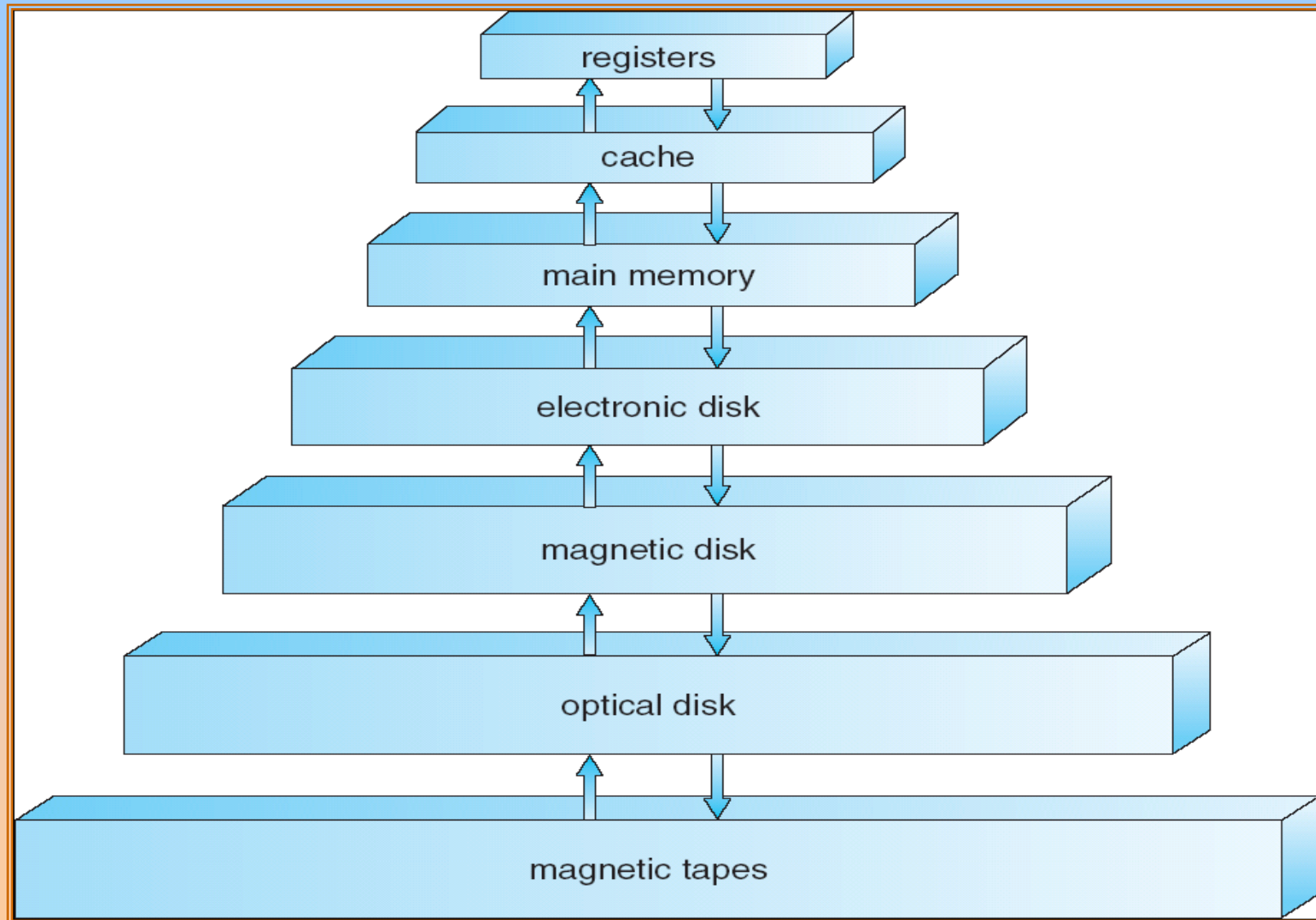
Storage Structure(存储结构)

- Main memory(内存) – only large storage media that the CPU can access directly.
- Secondary storage(外部存储) – extension of main memory that provides large nonvolatile storage capacity.
- Magnetic disks(硬盘) – rigid metal or glass platters covered with magnetic recording material
 - ☞ Disk surface is logically divided into *tracks*(磁道), which are subdivided into *sectors*(扇区).
 - ☞ The *disk controller* determines the logical interaction between the device and the computer.

Storage Hierarchy(存储层次结构)

- Storage systems organized in hierarchy.
 - ☞ Speed
 - ☞ Cost
 - ☞ Volatility
- *Caching(缓存)* – copying information into faster storage system; main memory can be viewed as a last *cache* for secondary storage.

Storage-Device Hierarchy



Computer-System Architecture

- Single-processor Systems(单处理器操作系统)
- MultiSingle-processor Systems(多处理器操作系统)
- Clustered Systems(集群系统)

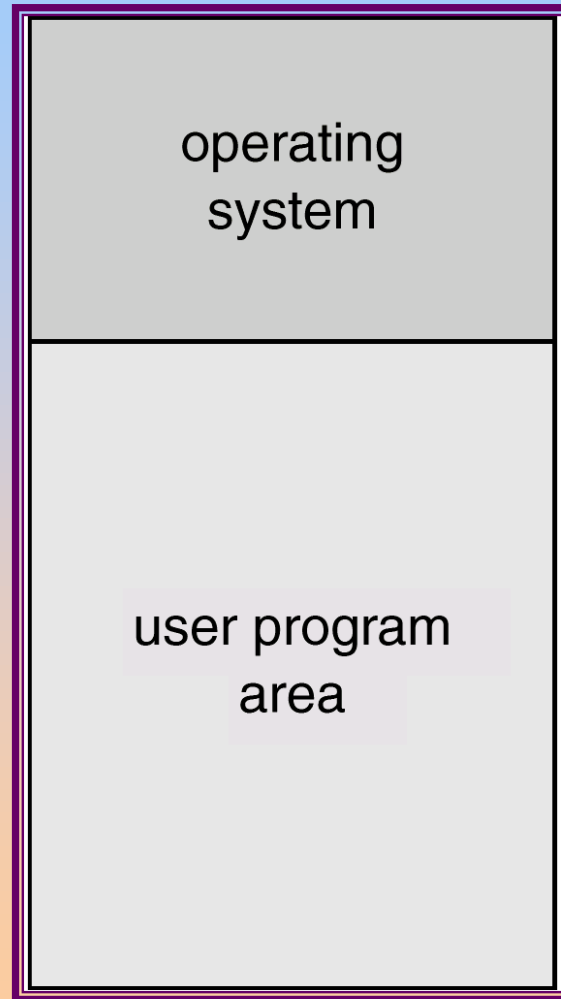
Operating System Structure

Multiprogramming(多道程序设计) needed for efficiency

- ➡ Single user cannot keep CPU and I/O devices busy at all times
- ➡ Multiprogramming organizes jobs (code and data) so CPU always has one to execute
- ➡ A subset of total jobs in system is kept in memory
- ➡ One job selected and run via **job scheduling**(作业调度)
- ➡ When it has to wait (for I/O for example), OS switches to another job

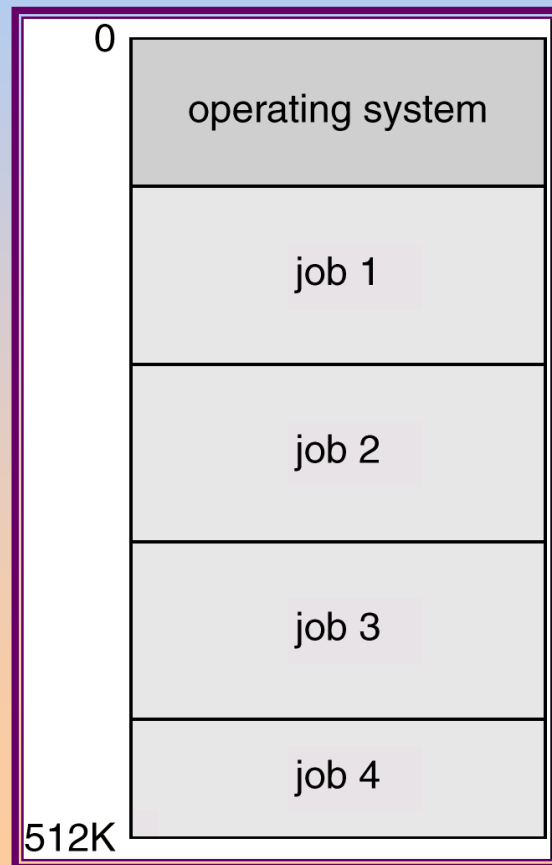
Timesharing (multitasking) is logical extension in which CPU switches jobs so frequently that users can interact with each job while it is running, creating **interactive** computing

Memory Layout for a Simple Batch System



Multiprogrammed Batch Systems

Several jobs are kept in main memory at the same time, and the CPU is multiplexed among them.



OS Features Needed for Multiprogramming

- I/O routine supplied by the system.
- Memory management – the system must allocate the memory to several jobs.
- CPU scheduling – the system must choose among several jobs ready to run.
- Allocation of devices.

Operating-System Operations

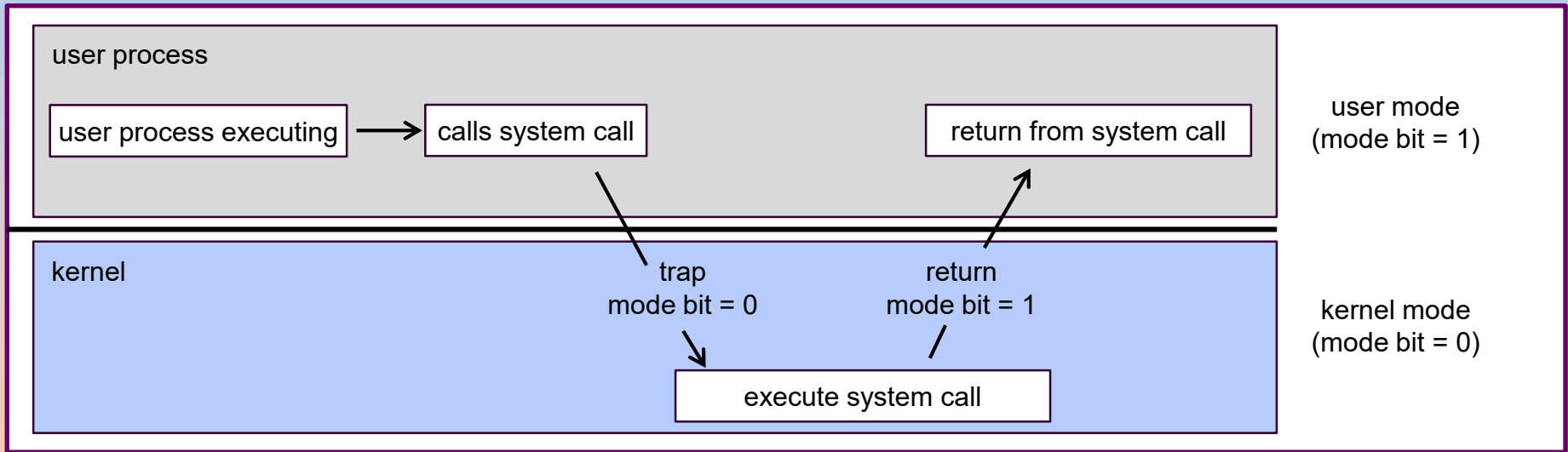
- Interrupt driven by hardware
- Software error or request creates **exception**(异常) or **trap**
 - ☞ Division by zero, request for operating system service
- Other process problems include infinite loop, processes modifying each other or the operating system

Operating-System Operations

- **Dual-mode(双重模式)** operation allows OS to protect itself and other system components
 - ☞ **User mode(用户模式)** and **kernel mode(内核模式)**
 - ☞ **Mode bit(模式位)** provided by hardware
 - 📄 Provides ability to distinguish when system is running user code or kernel code
 - 📄 Some instructions designated as **privileged(特权)**, only executable in kernel mode
 - 📄 System call changes mode to kernel, return from call resets it to user

Transition from User to Kernel Mode

- Timer to prevent infinite loop / process hogging resources
 - ☞ Set interrupt after specific period
 - ☞ Operating system decrements counter
 - ☞ When counter zero generate an interrupt
 - ☞ Set up before scheduling process to regain control or terminate program that exceeds allotted time



Process Management(进程管理)

- A process is a program in execution. It is a unit of work within the system. Program is a *passive entity*, process is an *active entity*.
- Process needs resources to accomplish its task
 - ☞ CPU, memory, I/O, files
 - ☞ data
- Process termination requires reclaim of any reusable resources
- Single-threaded process has one **program counter**(程序计数器) specifying location of next instruction to execute
- Multi-threaded process has one program counter per thread
- Typically system has many processes, some user, some operating system running concurrently on one or more CPUs

Process Management Activities

The operating system is responsible for the following activities in connection with process management:

- Creating and deleting both user and system processes
- Suspending and resuming processes
- Providing mechanisms for process synchronization
- Providing mechanisms for process communication
- Providing mechanisms for deadlock handling

Memory Management(内存管理)

- All data in memory before and after processing
- All instructions in memory in order to execute
- Memory management determines what is in memory when
 - ☞ Optimizing CPU utilization and computer response to users
- Memory management activities
 - ☞ Keeping track of which parts of memory are currently being used and by whom
 - ☞ Deciding which processes (or parts thereof) and data to move into and out of memory
 - ☞ Allocating and deallocating memory space as needed

Storage Management(存储管理)

- OS provides uniform, logical view of information storage
 - ☞ Abstracts physical properties to logical storage unit - **file(文件)**
 - ☞ Each medium is controlled by device (i.e., disk drive, tape drive)
 - 📄 Varying properties include access speed, capacity, data-transfer rate, access method (sequential or random)

■ File-System management

- ☞ Files usually organized into directories
- ☞ Access control on most systems to determine who can access what
- ☞ OS activities include
 - 📄 Creating and deleting files and directories
 - 📄 Primitives to manipulate files and dirs
 - 📄 Mapping files onto secondary storage
 - 📄 Backup files onto stable (non-volatile) storage media

Mass-Storage Management

(大容量存储器管理)

- Usually disks used to store data that does not fit in main memory or data that must be kept for a “long” period of time.
- Proper management is of central importance
- Entire speed of computer operation hinges on disk subsystem and its algorithms
- OS activities
 - ☞ Free-space management
 - ☞ Storage allocation
 - ☞ Disk scheduling

■ Some storage need not be fast

- ☞ Tertiary storage includes optical storage, magnetic tape
- ☞ Still must be managed
- ☞ Varies between WORM (write-once, read-many-times) and RW (read-write)

I/O Subsystem(I/O子系统)

- One purpose of OS is to hide peculiarities of hardware devices from the user
- I/O subsystem responsible for
 - ☞ Memory management of I/O including buffering (storing data temporarily while it is being transferred), caching (storing parts of data in faster storage for performance), spooling (the overlapping of output of one job with input of other jobs)
 - ☞ General device-driver interface
 - ☞ Drivers for specific hardware devices

Protection and Security

- **Protection(保护)** – any mechanism for controlling access of processes or users to resources defined by the OS
- **Security(安全)** – defense of the system against internal and external attacks
 - ☞ Huge range, including denial-of-service, worms, viruses,

- Systems generally first distinguish among users, to determine who can do what
 - ☞ User identities (**user IDs**(**用户ID**), security IDs) include name and associated number, one per user
 - ☞ User ID then associated with all files, processes of that user to determine access control
 - ☞ Group identifier (**group ID**) allows set of users to be defined and controls managed, then also associated with each process, file
 - ☞ **Privilege escalation** allows user to change to effective ID with more rights

Time-Sharing Systems–Interactive Computing

- The CPU is multiplexed among several jobs that are kept in memory and on disk (the CPU is allocated to a job only if the job is in memory, time slot(时间片)).
- A job swapped in and out of memory to the disk.
- On-line communication between the user and the system is provided; when the operating system finishes the execution of one command, it seeks the next “control statement” from the user’s keyboard.
- On-line system must be available for users to access data and code.

Desktop Systems

- *Personal computers* – computer system dedicated to a single user.
- I/O devices – keyboards, mice, display screens, small printers.
- User convenience and responsiveness.
- Can adopt technology developed for larger operating system' often individuals have sole use of computer and do not need advanced CPU utilization of protection features.
- May run several different types of operating systems (Windows, MacOS, UNIX, Linux)

Parallel Systems(并行系统)

- Multiprocessor systems with more than one CPU in close communication.
- *Tightly coupled system* – processors share memory and a clock; communication usually takes place through the shared memory.
- Advantages of parallel system:
 - ☞ Increased *throughput*(吞吐量)
 - ☞ Economical
 - ☞ Increased reliability

Parallel Systems (Cont.)

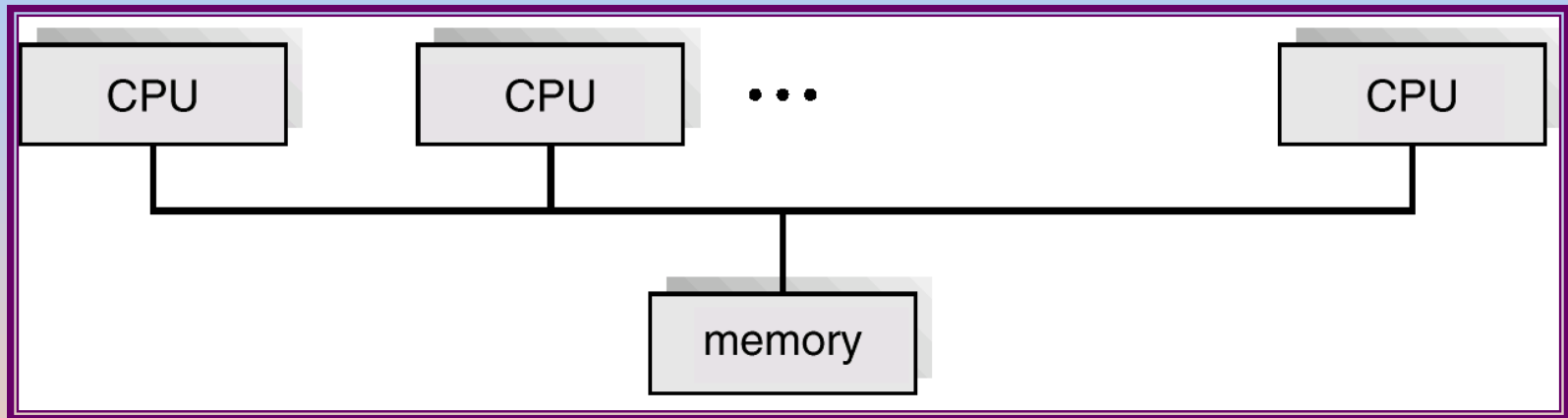
■ *Symmetric multiprocessing (SMP)*

- ☞ Each processor runs identical copy of the operating system.
- ☞ Many processes can run at once without performance deterioration.
- ☞ Most modern operating systems support SMP

■ *Asymmetric multiprocessing*

- ☞ Each processor is assigned a specific task; master processor schedules and allocated work to slave processors.
- ☞ More common in extremely large systems

Symmetric Multiprocessing Architecture



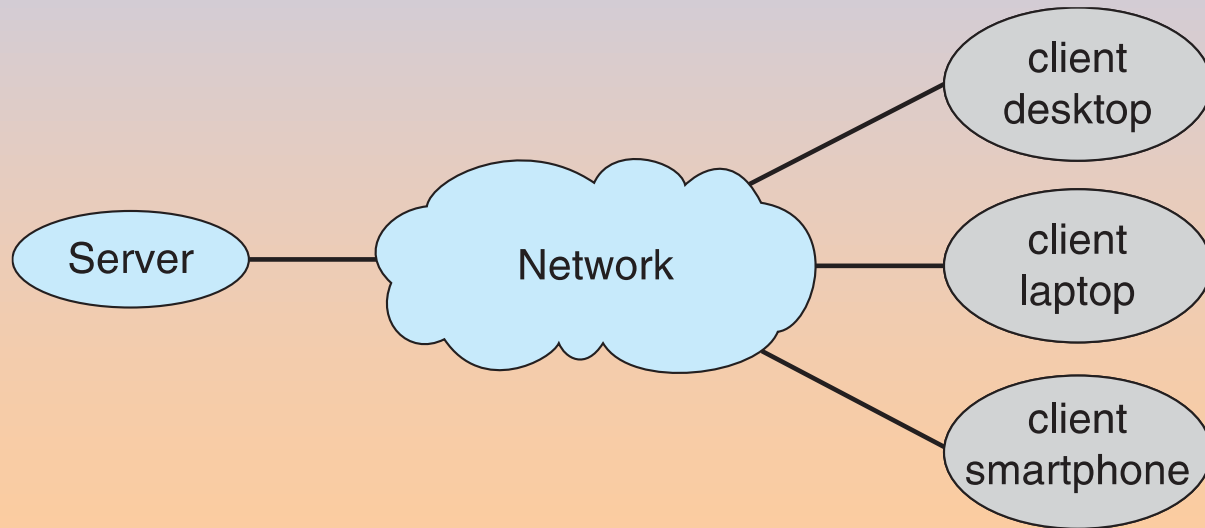
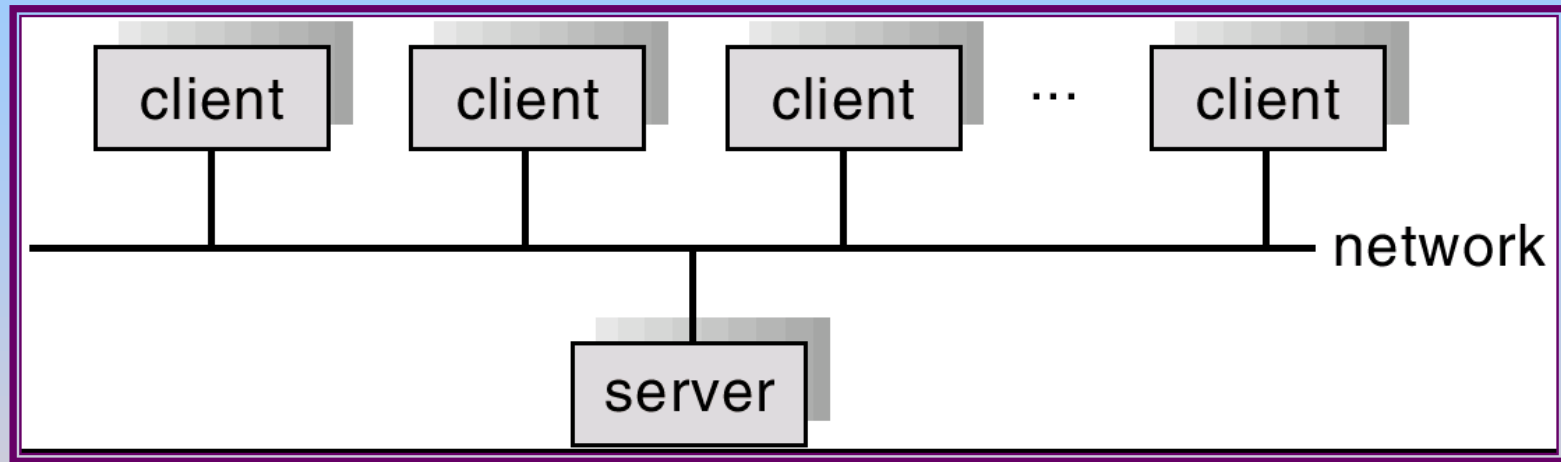
Distributed Systems

- Distribute the computation among several physical processors.
- *Loosely coupled system*(松耦合) – each processor has its own local memory; processors communicate with one another through various communications lines, such as high-speed buses or telephone lines.
- Advantages of distributed systems.
 - ☞ Resources Sharing
 - ☞ Computation speed up – load sharing
 - ☞ Reliability
 - ☞ Communications

Distributed Systems (cont)

- Requires networking infrastructure.
- Local area networks (LAN) or Wide area networks (WAN)
- May be either client-server or peer-to-peer (对等) systems.

General Structure of Client-Server



Clustered Systems(集群系统)

- Clustering allows two or more systems to share storage.
- Provides high reliability.
- *Asymmetric clustering(非对称集群)*: one server runs the application while other servers standby.
- *Symmetric clustering(对称集群)*: all N hosts are running the application.

Real-Time Systems(实时操作系统)

- Often used as a control device in a dedicated application such as controlling scientific experiments, medical imaging systems, industrial control systems, and some display systems.
- Well-defined fixed-time constraints.
- Real-Time systems may be either *hard* or *soft* real-time.

Real-Time Systems (Cont.)

■ Hard real-time:

- ☞ Secondary storage limited or absent, data stored in short term memory, or read-only memory (ROM)
- ☞ Conflicts with time-sharing systems, not supported by general-purpose operating systems.

■ Soft real-time

- ☞ Limited utility in industrial control of robotics
- ☞ Useful in applications (multimedia, virtual reality) requiring advanced operating-system features.

Handheld Systems

- Personal Digital Assistants (PDAs)
- Cellular telephones
- Issues:
 - ☞ Limited memory
 - ☞ Slow processors
 - ☞ Small display screens.

Operating System Evolution

- In the ten years since the birth of computer:
 - ☞ Computers are regarded by most people as special machines for fixed computing work;
 - ☞ Add operating system: consume more storage and computing resources and reduce efficiency;

Operating System Evolution (Cont.)

- The emergence of OS / 360 operating system:
 - ☞ Make it easier for users to write and manage programs in advanced languages;
 - ☞ Applications that can be upgraded to hardware;
 - ☞ Facilitate the development, use and maintenance of applications.
- IBM决定为其第一款多功能大型机 S/360 开发配备后来软件工程领域颇具传奇色彩的 OS/360 操作系统。

Operating System Evolution (Cont.)

- In the 1970s, the development of integrated circuit technology:
 - ☞ Computers began to develop towards miniaturization and miniaturization;
 - ☞ PDP-7 gave birth to a new generation of operating system UNIX.

Operating System Evolution (Cont.)

- UNIX and open source movement:
 - ☞ Modular design and pipeline connection between processes are adopted;
 - ☞ The design follows the principle of cognitive engineering;
 - ☞ It gradually evolved into two versions: BSD and system V.

Operating System Evolution (Cont.)

■ Open source movement and Linux:

- ☞ In the 1980s, free software flourished with the popularity of personal computers.
- ☞ The purpose is to let more people use open source software;
- ☞ A series of agreements and license systems with different degrees of source code control have been formed;

Operating System Evolution (Cont.)

■ Windows:

- ☞ Serving the PC market;
- ☞ Become the first computer operating system familiar to the public;
- ☞ Dominate the whole market of personal computer operating system in the 1990s;
- ☞ Due to the defects of system design, the system is easy to be attacked by hackers;

Operating System Evolution (Cont.)

- In the 1970s, gave birth to a new generation of operating system UNIX.
- UNIX and open source movement
- Windows: serving the PC market, Microsoft Windows became the computer operating system familiar to the public, and dominated the entire PC operating system market in the 1990s.

Operating System Evolution (Cont.)

■ Apple and Android:

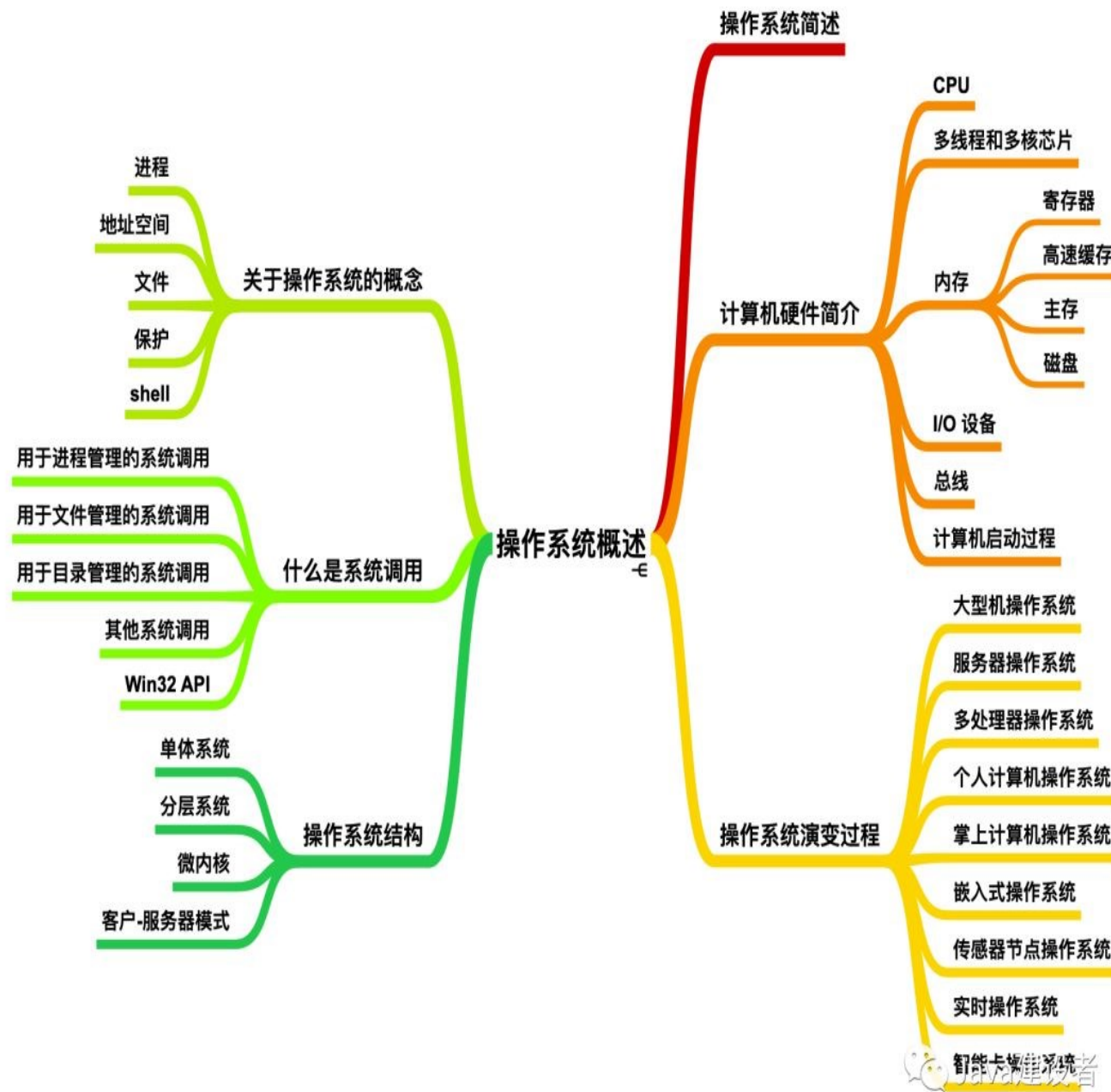
- ☞ Mobile computing is a new stage of computer development after personal computer.

Operating System Evolution (Cont.)

- Open source movement and Linux: in the 1980s, free software flourished with the popularity of personal computers.
- AIX (Advanced Interactive eXecutive)
HP-UX (Hewlett-Packard UNIX)
Solaris
- Apple and Android
- openEuler, **Harmony OS**(鸿蒙系统), 麒麟操作系统

Computing Environments

- Traditional computing
- Web-Based Computing
- Embedded Computing



引言

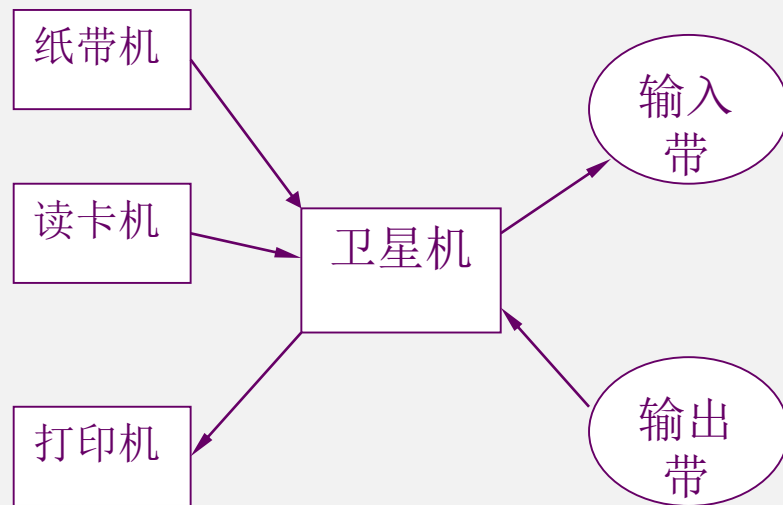
- 操作系统（Operating System，简称OS）传统上是负责对电脑硬件直接控制及管理的系统软件。
- 对计算机系统而言，操作系统是对所有系统资源进行管理的程序的集合；
- 对用户而言，操作系统提供了对系统资源进行有效利用的简单抽象的方法。

引言

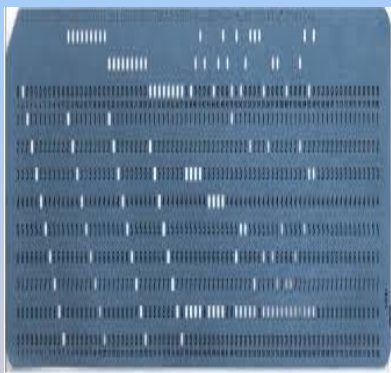
- 操作系统（Operating System，简称OS）是计算机系统中的一个系统软件，管理和控制计算机中的软件和硬件资源，用户提供一个功能强、使用方便的工作环境，从而在计算机与其用户之间起到接口的作用。
- 对计算机系统而言, 操作系统是对所有系统资源进行管理;
- 对用户而言, 操作系统提供了 系统接口, 方便用户使用计算机。

手工操作阶段

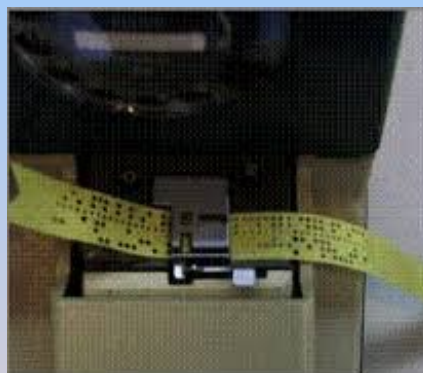
- 人工把源程序用穿孔机穿制在卡片或纸带上;
- 将准备好的汇编解释程序或编译系统装入计算机;
- 汇编程序或编译系统读入人工装在输入机上的穿孔卡或穿孔带;
- 执行汇编过程或编译过程, 产生目标程序, 并输出目标卡片或纸带;
- 通过引导程序把装在输入机上的目标程序读入计算机;
- 启动目标程序执行, 从输入机上读入人工装好的数据卡或数据带;
- 产生计算结果, 执行结果从打印机上或卡片机上输出。



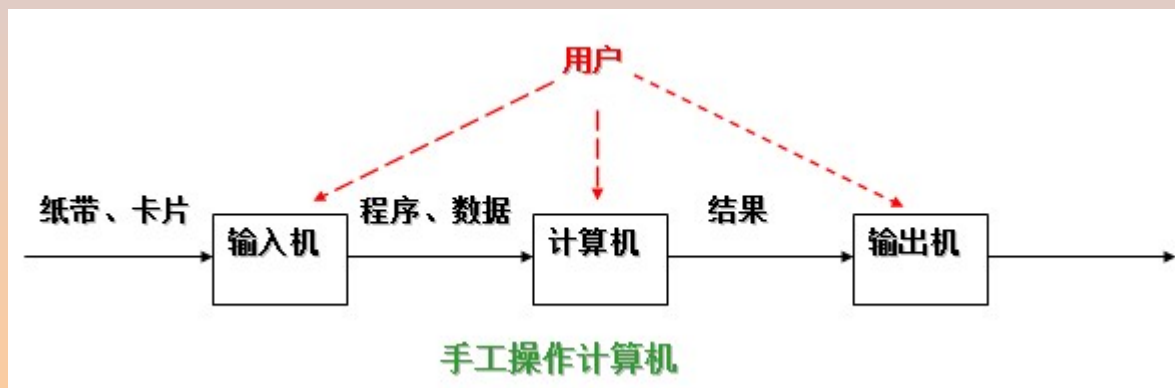
- 早期，计算机是没有操作系统的
- 随着计算机的发展，计算机系统的硬件和软件资源也越来越丰富了。为了提高这些资源利用率和增强系统的处理能力，出现了监督程序，作为用户与计算机间的接口。
- 到了60年代，监督程序又进一步发展为操作系统。



卡片punch card



打孔纸带



批处理系统

- 在20世纪50年代，计算机的价格非常昂贵，为了想办法减少机时的浪费，人们开发了批处理系统
- 用一台较便宜的机器将输入的数据读到磁带上，收集批处理作业
- 用一台较昂贵的机器来进行计算
- 批处理系统，在它的控制下，计算机能够自动地、成批地处理用户的作业

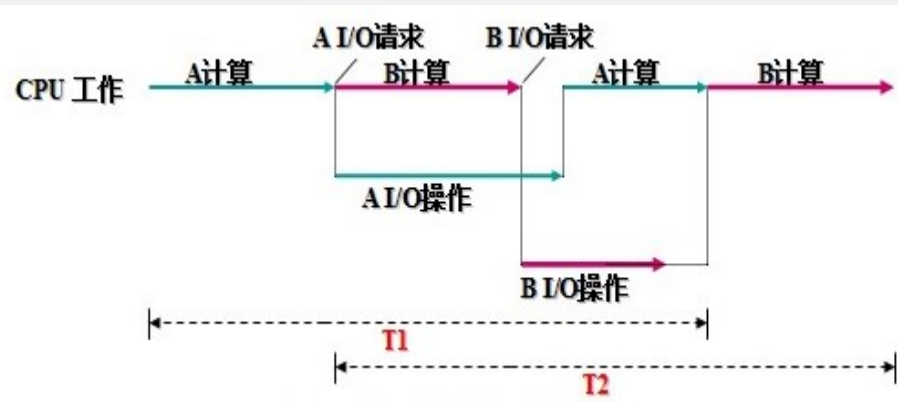
批处理系统

- 批处理系统一次只能运行一个程序
 - 当程序在I/O时，CPU是空闲的
 - 当程序在计算时，I/O系统是空闲的
- 为了解决批处理系统一次只能运行一个程序的问题，人们开发了多道程序系统
 - 多道程序设计技术允许多个程序同时进入内存并运行，并允许它们交替在CPU中运行，它们共享系统中的各种硬、软件资源
- IBM OS/360操作系统是典型的多道程序技术
- 多道程序系统通常也被认为是多进程系统
- 多道程序系统、分时系统的出现，标志着现代 操作系统 的出现

多道程序系统



批处理系统工作示例



多道程序工作示例

多线程

- 传统多进程系统存在以下缺点
 - 占用内存多
 - 进程间的切换复杂
 - CPU利用率低
- 为了进一步节省系统的内存资源和帮助任务间的通信，20世纪80年代中期，Solaris将多线程技术应用到操作系统中
 - 线程节省系统内存，CPU利用率高
 - 线程间共享数据，方便通讯和切换
- 多线程技术被广泛应用在当前的主流操作系统中，如Unix，Linux和Windows等

- 随着CPU速度不断提高，为了解决多用户同时使用计算机的问题，在1962年，麻省理工学院在IBM 7094上开发出了第一个正式的分时系统CTTS (Compatible Time Sharing System)
- 分时系统的特点：
 - 人机交互性好：在调试和运行程序时由用户自己操作
 - 多用户同时性：多个用户同时使用
 - 用户独立性：每个用户好象在独占主机
- 多用户分时操作系统是当今计算机操作系统中使用最普遍的一类操作系统

实时处理系统

- 能在限定的时间内对输入进行快速处理并作出响应的计算机处理系统。
- 硬实时系统：主要用于工业生产的过程控制，武器的制导等。这类操作系统要求响应速度十分快，工作极其安全可靠。
- 软实时系统：对响应的速度要求不像硬实时系统那么高，且时限要求也不那么严的信息、事务处理领域，如航空、火车订票系统等。

智能卡操作系统

- 智能卡

- ✓ CPU芯片

- ✓ ROM

- 面向Java的智能卡

- ✓ JVM解释程序

- ✓ 下载Java applet并执行

- 嵌入在掌上电脑、手机、通讯设备、车载系统、信息家电等非计算机类设施上的操作系统。
- 特点：微内核结构（Micro-kernel），许多操作系统功能以应用程序模式运行。
- 例子：
 - ✓ Win CE .NET
 - ✓ PalmOS
 - ✓ HOPEN
 - ✓ Androd

Unix操作系统

- 1969年，美国贝尔实验室开发出了著名的多用户分时操作系统UNIX，主要开发人员有肯·汤普逊和丹尼斯·里奇等。
- Unix操作系统
 - Unix操作系统与开源操作系统
 - 多种版本：贝尔实验室的Unix System V和UC伯克利的FreeBSD
- 是一直沿用至今的商业服务器操作系统，商业应用需要收费。

Linux操作系统

- 虽然UNIX操作系统性能优越，但是由于UNIX操作系统对于商业应用是收费的，为了实现使用类似UNIX功能的免费开源操作系统，芬兰的Linux Torvalds在1991年编写了Linux操作系统
- Linux操作系统是当前广泛应用商业服务器的免费开源操作系统
 - 是一个支持多用户、多进程、多线程、实时性较好且稳定的操作系统

Windows操作系统

- 20世纪80年代初，主流的操作系统都是基于字符命令的，典型的有IBM-DOS和MS-DOS。字符界面不够友好增加了用户的操作难度
- Microsoft公司于1983年宣布推出Windows。
 - 1985年正式推出了Windows 1.0操作系统，其是基于DOS操作系统但提供图形化的用户界面
 - 1992年4月发布了Windows 3.1，该版本要求应用程序只能在保护模式下运行。DOS应用程序则在MSDOS窗口下运行，并可支持多任务。该版本支持虚拟存储方式，地址空间大幅扩张。
 - Windows 3.1是在中国推开的里程碑。

网络操作系统

- 在早期推出局域网时，同步出现了管理这些局域网的网络管理系统，如Novell公司推出了NetWare网络操作系统，用于提供网络通讯、网络资源共享（共享硬盘、打印机等）、网络服务的软件包。
主要采用客户端与服务器端（C/S结构）模式，支持无盘工作站的运行。
- 由于网络计算的出现和发展，现代操作系统已将本机操作与网络操作结合起来，故现在很少特指某操作系统为网络操作系统。
- 现代网络操作系统理念：
 - ✓ 软件服务化，计算虚拟化，位置透明化，交互普适化

虚拟机

- 由于早期计算系统资源昂贵，为了让用户能够充分利用大型机资源，虚拟机技术应运而生
- 1959年，提出了虚拟化的基本概念
- 1965年，IBM公司在“M44/44X”计算机项目中，定义了虚拟内存管理机制，用户程序可以运行在虚拟的内存中
- 由于x86架构本身不适合进行虚拟化，虚拟化技术一度进展缓慢
- 随着x86架构处理器性能的飞速提高，以及英特尔、AMD对x86处理器指令集的解决，x86架构上的虚拟化技术得到了比以前更大的关注。
- 代表厂商：VMware，Virtual Box

■ 多线程

- 传统多进程系统存在以下缺点
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 - 进程间的切换复杂
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- 为了进一步节省系统的内存资源和帮助任务间的通信，20世纪80年代中期，类Unix操作系统solaris将多线程技术应用到操作系统中
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- 能在限定的时间内对输入进行快速处理并作出响应的计算机处理系统。
- 硬实时系统：主要用于工业生产的过程控制，航天系统的跟踪和控制，武器的制导等。这类操作系统要求响应速度十分快，工作极其安全可靠。
- 软实时系统：对响应的速度要求不像硬实时系统那么高，且时限要求也不那么严密的信息查询、事务处理领域，如情报资料检索、订票系统、银行财务管理系统、信用卡记账取款系统、仓库管理系统等。

分布式操作系统

分布式操作系统负责管理分布式处理系统资源和控制分布式程序运行。

- ✓ 具有一个统一的操作系统。
- ✓ 它可以把一个大任务划分成很多可以并行执行的子任务，并按一定的策略将它们动态地分配给各个计算机执行。
- ✓ 可以控制管理各个计算机的资源分配、运行及计算机之间的通信，以协调任务的并行执行。

国产OS

■ 中标麒麟国产OS

- 2003年推出，中标软件（中国）设计
- 用户量：5000万左右（主要用户政府机构）

■ Alios 操作系统

- 2018年推出，阿里
- 以驱动万物智能为目标

■ Huawei HarmonyOS

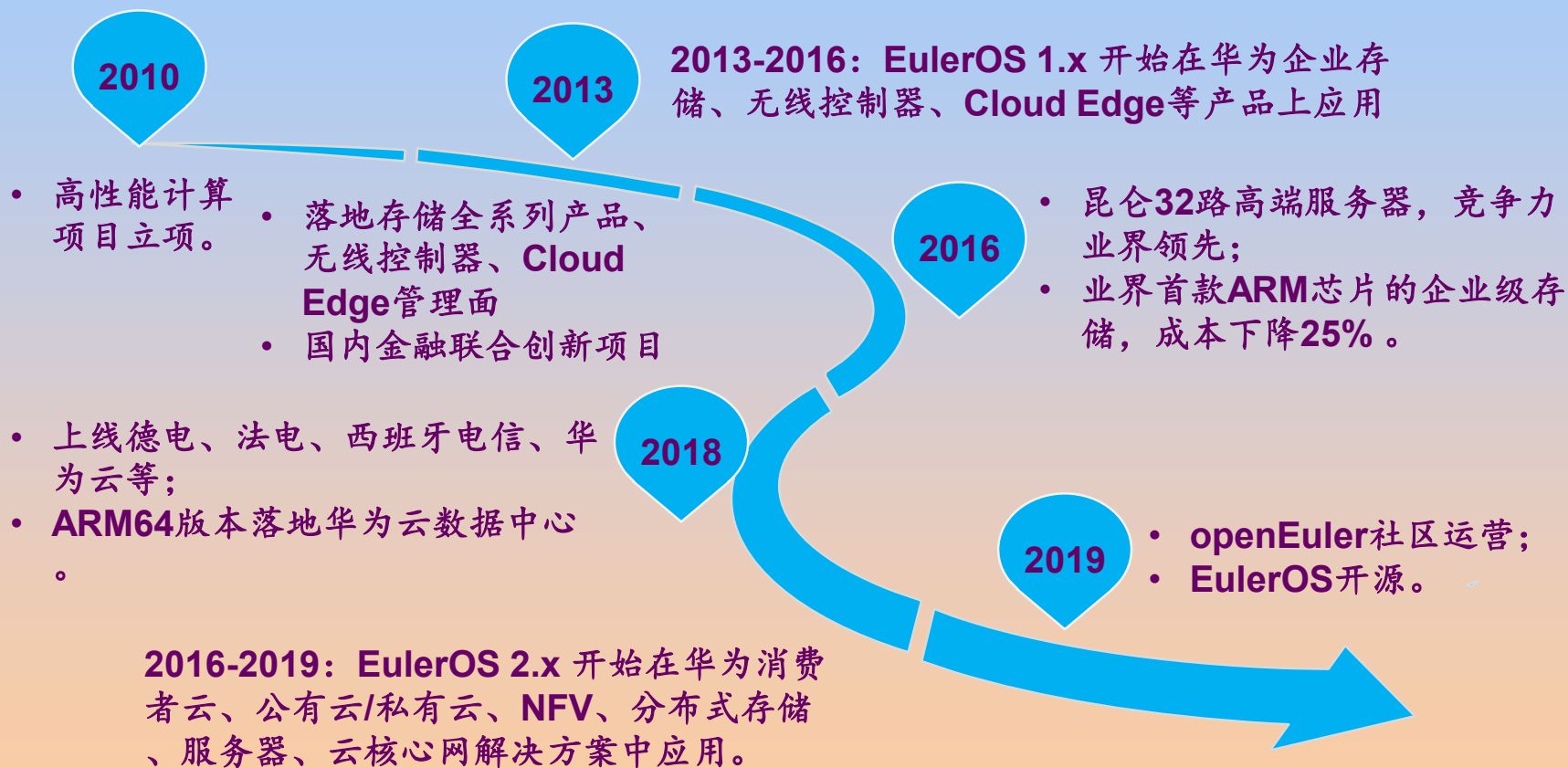
- 2019年8月推出，华为
- 基于微内核的面向全场景的分布式操作系统

■ UOS, “UnionTech OS”

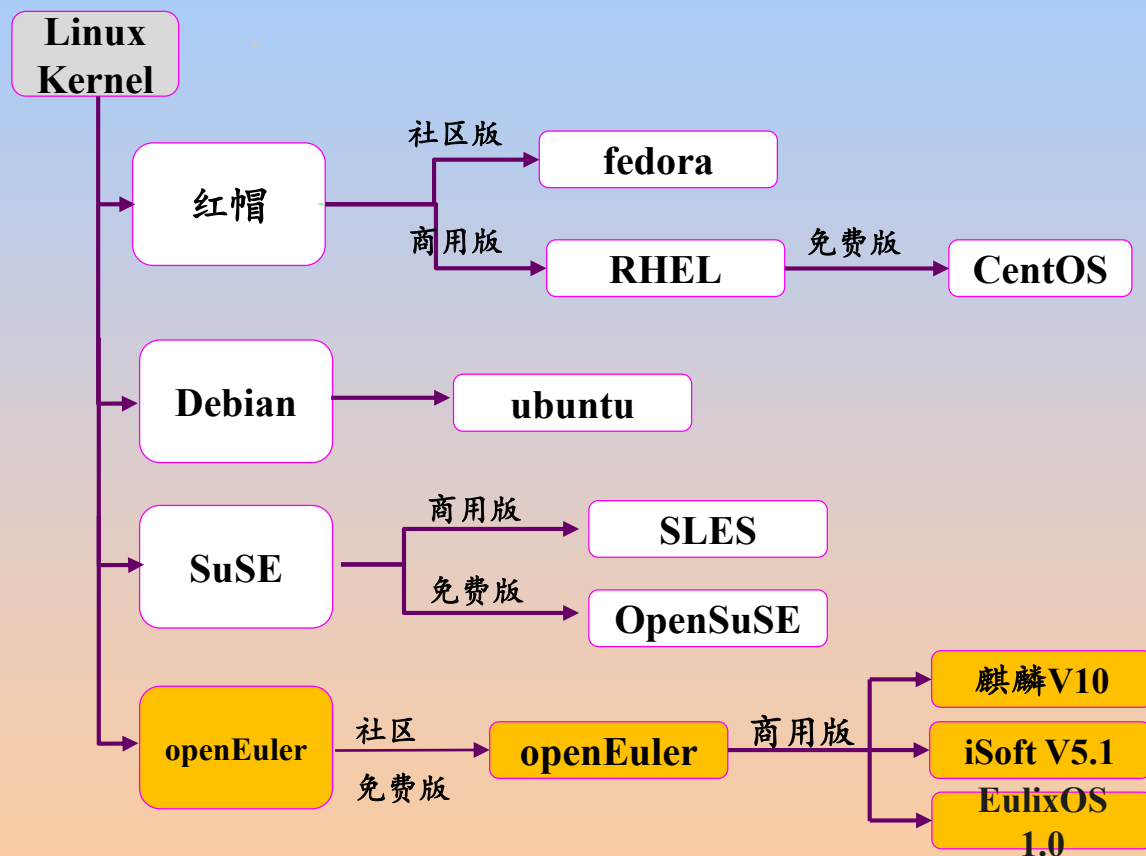
- ☞ 由统信软件开发的一款基于Linux内核的操作系统
- ☞ 2020年8月12，发布UOS 桌面操作系统 V20 专业版

■ 华为台式机即将登场，鲲鹏芯片+统信UOS

国产OS



国产OS



- openEuler与SuSE、Debian、RedHat的上游社区都是kernel社区 www.kernel.org
- openEuler社区发行 LTS免费版本，使能OSV发展商业发行版，如麒麟软件、普华、中科软、万里开源等

移动OS

- iOS
- Android
- vivo 的Hi-Fi& Smart （Funtouch OS）
- 小米操作系统（MIUI）
- 魅族操作系统（flyme OS）
- 锤子操作系统（Smartisan OS）
- 360 OS
- 华为操作系统（emui）
- HarmonyOS

